Kedros Analytics

DEMAND FORECASTING

Know What Will Sell Next Month

Turn weekly sales history into a forecast per product, an honest accuracy score and the right amount of safety stock.

Company Page · Blog

Matt Protopapas, PhD

THE OPPORTUNITY

The order goes in weeks before the sales happen

Every unit a gut feeling gets wrong becomes a stockout or a pallet of stock that sits until spring.

THE GUT FEELING

“About the same as last month.”

“We always do well on that one in October.”

Sometimes it is right. But the supplier’s lead time is fixed, so the mistake is locked in before anyone knows.

A NUMBER PER PRODUCT

What will sell, how sure you can be, and what the doubt should cost.

A forecast with a stated accuracy, linked directly to safety stock.

WHAT IT IS

From “about the same as last month” to a forecast you can score

Three textbook methods, an honest test, and a direct link from forecast error to safety stock.

1

Weekly sales

SKU, week and units sold for two to three years. A promotion flag helps most.

2

Three methods, scored blind

Seasonal naive, moving average and Holt-Winters. The last 13 weeks are held back.

3

Forecast, error, buffer

Next four weeks per product, with the safety stock its error implies.

The real output is not the forecast but the measurement of how wrong it is.

Demo: 30 SKUs · 156 weeks · 4,681 raw rows. Synthetic data.

WHAT YOU GAIN

Six ways it pays off for a small business

1 A forecast per product

Three methods turn weekly sales into a forecast, using only data you already have.

2 A fair test

The last quarter is held back and forecast blind, so methods are scored on your data.

3 An error you can plan with

MAE, MAPE and RMSE turn “roughly right” into “wrong by 12% in a typical week”.

4 Catches systematic mistakes

Bias shows whether a method over- or under-forecasts every single week.

5 Forecastable or not

Products are banded Easy, Moderate or Hard, so you know where to use judgement.

6 Feeds safety stock

A forecast that captures seasonality implies less buffer: that is the cash it buys.

Highlighted: the honest test and the link to inventory.

EXAMPLE RESULTS

Six in ten products are good enough to order from

Products banded by their best method's error on the 13 held-back weeks.

EASY · 3 SKUS

9.1%

Average error, 10% of products.

Under 15% error. Order from the forecast.

MODERATE · 18 SKUS

22.3%

Average error, 61% of volume.

15–30% error. Order with a sensible safety stock.

HARD · 9 SKUS

40.2%

Average error, 24% of volume.

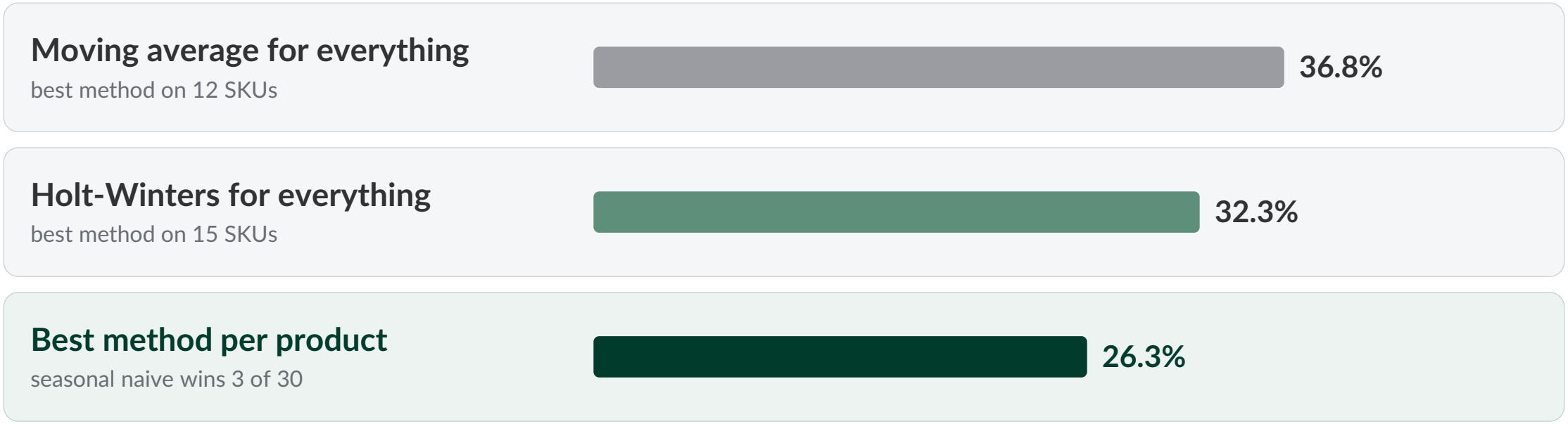
Over 30% error. A starting point, not an answer.

The bands are a judgement, not a law: 15% and 30% are settings you can change.

WHICH METHOD

The cleverest method wins less than half the time

Average holdout error across all 30 SKUs, by how the method is chosen.



Pick per product: 6 points less error than the best single method.

Synthetic demo data. Bars scaled to average error (MAPE).

MATCHING METHOD TO PRODUCT

The winner matches the product’s behaviour

Holdout error (MAPE) for three products from the league table.

SKU-025

5.0%

Moving average wins.

Demand had stepped down. Holt-Winters (24.3%) and seasonal naive (18.9%) kept forecasting last year’s level.

SKU-001

11.7%

Holt-Winters wins.

A clear seasonal pattern. Moving average 12.4%, seasonal naive 16.4%.

SKU-030

109.6%

Moving average fails.

One promotion week of 534 units inflated its window: +106 units a week of bias. Seasonal naive scored 25.1%.

Synthetic demo data. SKU-025 and SKU-001 show the best method’s error; SKU-030 the moving average’s.

READ THE BIAS

A good error score can hide a systematic miss

Bias is the signed error: does the forecast run high or low, week after week?

SKU-012

+22.4%

Over-ordered by a fifth.

A 28.2% MAPE looks acceptable. The moving average would over-order every week of the quarter.

SKU-022

−24.3%

A stockout waiting to happen.

The mirror image: the forecast runs consistently below actual sales.

SKU-020

−17.9%

Despite a 15.4% MAPE.

One promotion week sold 830 units against a forecast of 287.

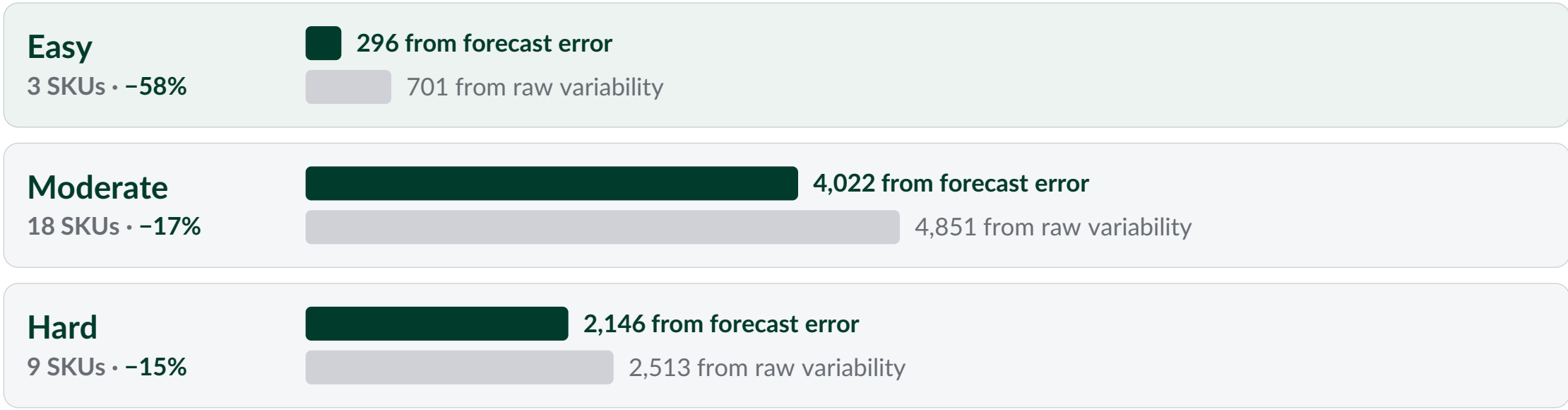
Promotion weeks were forecast twice as badly: 48.7% error against 25.1% for ordinary weeks.

Synthetic demo data. Bias shown as a percentage of demand.

FROM ERROR TO SAFETY STOCK

Sizing the buffer on forecast error releases cash

Safety stock units by forecastability band, against the classic formula that ignores the forecast.



All 30 products: 6,464 units against 8,065, about €46,000 less inventory at list price.

Synthetic demo data. Bars scaled to units.

Kedros Analytics

BUSINESS IMPACT

A better forecast is worth working capital

Demo distributor: the next four weeks forecast at roughly €1.02 million at list price.

€46,000

less safety stock, about one-eighth of the buffer, without touching service levels.

6 points less error

26.3% choosing per product, against 32.3% for the best single method.

7 products with a clear bias

Three over-ordered by about a fifth, four under-ordered by a fifth to a half.

Synthetic demo data. List-price valuation.

Kedros Analytics

GET STARTED

What to do this week

Five steps from a sales export to next month’s order.

1

Export

Weekly sales by SKU for two to three years. Add a promotion flag if you can.

2

Hold back

Keep the last quarter hidden and score each method on it before trusting it.

3

Read

Start with forecastability, then the bias column for every SKU you are about to order.

4

Recompute

Safety stock from forecast error, against what you hold today.

5

Repeat

Re-run monthly. A product whose best method flips often is changing.

Where the forecast-based buffer is higher, ask what the forecast is missing: usually promotions or lumpy customers.

Open-source demo: github.com/MattProtopapas/simple-demand-forecasting

KEEP IT HONEST

The limits worth keeping in mind

The forecast is a baseline for a human to adjust, not a prediction of the future.

1 Only knows the history

New products, lost customers and next month’s promotion are invisible. A human adjusts the baseline.

2 One quarter is one sample

Thirteen weeks tells Easy from Hard. It cannot separate 15.4% from 16.2%.

3 MAPE breaks on small numbers

For lumpy movers, read units and zero-sales weeks, and use min/max rules.

4 Seasonality needs history

Holt-Winters needs two full seasons, and learns each week from only three examples.

HOW KEDROS ANALYTICS CAN HELP

Try it on your own data, with a free proof of concept

Business questions first, AI as a tool and not a hype machine, and results you can act on straight away.

1

Validate

Check your weekly sales export for negative values, bad dates, unknown codes and gaps.

2

Forecast and score

Test several methods blind on your own history and pick the best per product.

3

Order and track

Turn the forecast into order quantities and safety stock, then watch accuracy over time.

Free PoC: tailored to your requirements and your company’s analytics maturity.

Kedros Analytics

Ready to see how well your demand can be forecast?

Let’s spend thirty minutes on your sales history. No commitment, no jargon, and we might even plan a free PoC.

Book a free call

Matt@KedrosAnalytics.com



Company Page · Blog

Matt Protopapas, PhD · Founder